

REFERRING MULTI-OBJECT TRACKING IN MOVING-CAMERA VIDEOS VIA GLOBAL MOTION COMPENSATION

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ABSTRACT

Referring multi-object tracking (RMOT) takes a video and a language expression as input and tracks all referred objects. Many referring expressions describe how an object moves rather than how it appears. However, in a video captured by a moving camera, a parked vehicle may appear to move, while a moving vehicle may show little displacement. Existing RMOT methods relate motion to text but do not explicitly remove camera-induced motion. In this paper, we propose to estimate camera motion in driver-view videos, extract the residual object motion across frames, and compare its characteristics with the query expression. We score how well an object’s residual motion matches the expression and fuse that score with the host RMOT method’s prediction by late fusion. On the Refer-KITTI benchmarks, plugging the module into three RMOT architectures (iKUN, TransRMOT, and FlexHook) improves pooled HOTA in all four host settings and raises moving-class HOTA on iKUN by 9.44 points, at 6.7 ms per frame on a CPU and without retraining the host.

Index Terms— referring multi-object tracking, motion compensation, motion-language alignment

1. INTRODUCTION

Referring multi-object tracking (RMOT) outputs tracking results of the objects described by a free-form language expression. For example, a description such as “left vehicles in red” targets the red vehicles on the left. However, many descriptions focus on moving objects, such as “moving cars” or “a vehicle turning left”. In videos captured by a moving camera, the observed motion is a mixture of object motion and camera motion; thus, a static car may appear to move, while a moving car may appear static across frames.

Previous RMOT methods [1, 2, 3] match language expressions to objects. VMRMOT [4] considers explicit motion cues, such as changes in direction and speed, in the matching process. These methods compute motion from the object’s bounding-box displacement across frames, which mixes object motion with camera motion. In contrast, conventional multi-object tracking (MOT) methods explicitly remove camera motion: BoT-SORT [5] estimates camera-induced image motion, while UCMCTrack [6] maps detections to a ground plane to compensate camera motion. These methods, however, stop at tracking and do not connect the compensated motion to language.

In this paper, we propose a plug-in module that separates object motion from camera motion and provides motion cues for RMOT methods. The module consists of four stages. First, the geometric transformation across frames is estimated to measure camera motion. The estimated transformations are composed across frames so that camera motion can be measured over longer temporal spans. Second, residual velocity, which represents object motion after removing camera motion, is computed across frames and used as a temporal motion feature. Third, a contrastive motion–language aligner is trained to

compare motion features with language expressions and produce a motion-matching score. Finally, the motion-matching score is scaled by a host-specific weight and added to the host’s matching score. Here, the host model refers to an existing RMOT method used as a baseline, namely iKUN [2], FlexHook [3], and TransRMOT [1] in our experiments. The proposed process is implemented as a plug-in module, thereby improving existing RMOT baselines without retraining them.

This work connects motion compensation with motion-language modeling for RMOT. Our main contributions are as follows.

- We propose a plug-in module that aligns the object’s residual velocity with the referring expression, thereby improving the host RMOT method without modifying or retraining it.
- Across four host settings spanning three RMOT architectures, i.e., iKUN, TransRMOT, and FlexHook on two benchmarks, the proposed module improves pooled HOTA in every setting. On iKUN, it raises moving-class HOTA from 27.70 to 37.14 ± 0.92 (+9.44 points).
- We observe that the moving-class gain grows as the host’s own motion understanding weakens: the weakest host gains the most and the strongest the least, consistently across all four settings.

2. RELATED WORK

2.1. Referring Multi-Object Tracking

TransRMOT [1] introduced the RMOT task on the Refer-KITTI dataset. Subsequent work has improved individual components of this pipeline. iKUN [2] separated language-based object matching from object tracking, while FlexHook [3] redesigned the feature construction and cross-modal correspondence of the two-stage pipeline, handling motion implicitly through temporal feature differences. ReferGPT [7] used a multimodal large language model to match object tracks with expressions in a zero-shot manner, and MEX [8] provided a memory-efficient module for matching tracks with language. TempRMOT [9] extends TransRMOT with a temporal query memory and introduces the Refer-KITTI V2 benchmark.

A recent method [10] further improves how object descriptions are understood. STORM [11] unifies grounding and tracking in an end-to-end multimodal large language model. VMRMOT [4] explicitly represents motion using information such as position, direction, changes in distance, and changes in speed. Like VMRMOT, these methods derive object motion from bounding-box displacement in the image plane, which conflates object motion with camera motion; to our knowledge, none applies camera-motion compensation.

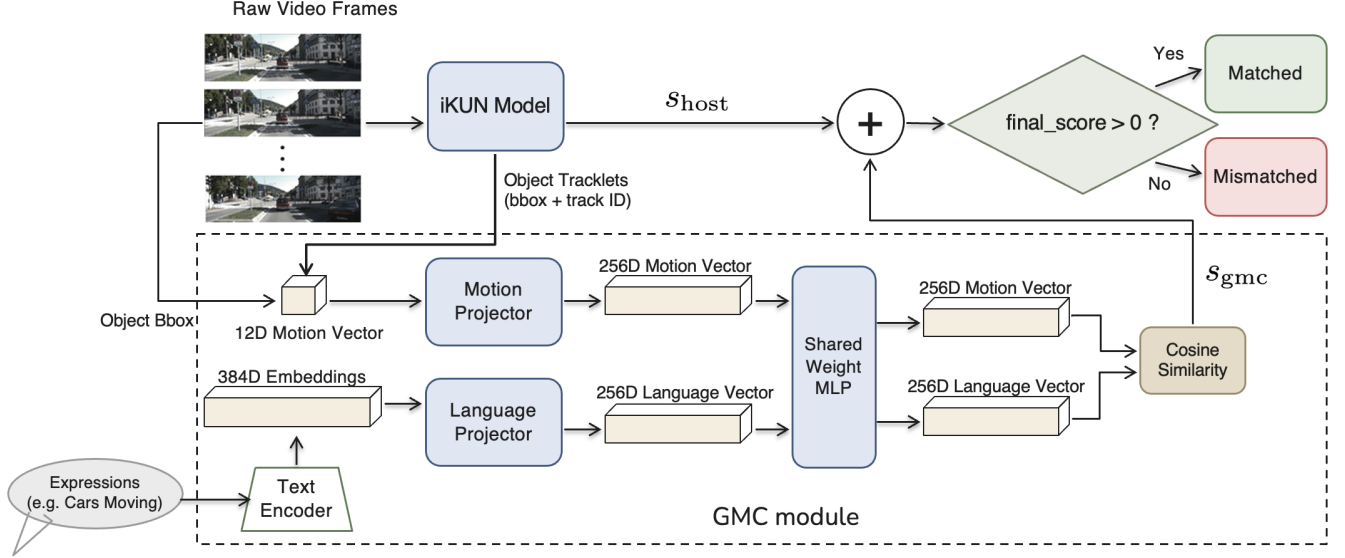


Fig. 1. The proposed RMOT framework with the global motion compensation (GMC) module. The host model (iKUN in this case) predicts an object-language matching score. The GMC module provides an extra matching score to adjust matching results.

2.2. Motion Compensation in Tracking

Camera motion compensation is widely used in conventional multi-object tracking. BoT-SORT [5] estimates camera-induced motion across frames. UCMCTrack [6] performs this compensation on a ground plane. EMAP [12] further separates the camera’s rotation and translation from object motion within the Kalman prediction process.

These methods demonstrate that compensating for camera motion can improve tracking. However, how to integrate compensated motion cues with language expressions to improve RMOT remains underexplored.

3. METHOD

Figure 1 shows the proposed RMOT framework with the global motion compensation (GMC) module. The GMC module is attached to a host RMOT method, iKUN [2] in this case. iKUN takes the tracklets produced by a frozen off-the-shelf detector and tracker and outputs an object-language matching score for each of them. The inputs to the GMC module are the sequence of object bounding boxes across frames, the video frames, and the language expression. Based on these inputs, it estimates camera motion, extracts residual object motion, aligns that residual motion with the language expression, and produces another matching score. The score is then combined with the host model’s predicted score to determine the final tracking result. The GMC module consists of four stages: motion compensation, multi-scale residual velocity extraction, motion-language matching, and fusion.

3.1. Motion Compensation

This stage follows the sparse-feature camera-motion-compensation recipe of modern trackers such as BoT-SORT [5] — sparse keypoints, optical flow, and robust homography fitting with the tracked object boxes masked out — but restricts the correspondences to the road region. As in previous RMOT works, we take Refer-KITTI [1], built on KITTI [13], as the evaluation benchmark. Refer-KITTI is recorded

from a driver-view camera, so the road surface is visible in most frames and stays close to a single plane. For a forward-facing camera, the horizon lies near the vertical image center, so the lower half of the frame is dominated by the road surface. We extract Shi–Tomasi keypoints [14] in this lower half, excluding the interiors of the tracked object boxes, and track them into the next frame with pyramidal Lucas–Kanade optical flow [15]. RANSAC [16] is then applied to these correspondences to estimate the frame-to-frame homography $H_{t-1 \rightarrow t}$. Although the correspondences are sampled locally, the estimated homography serves as the global frame-to-frame camera-motion model: full-frame keypoint sets place most correspondences off the road plane, which degrades the fit for exactly the near-ground region where the referred objects move.

To describe camera motion across more frames, the estimated homographies are accumulated as

$$H_{t-k \rightarrow t} = H_{t-1 \rightarrow t} \times H_{t-2 \rightarrow t-1} \times \cdots \times H_{t-k \rightarrow t-k+1}, \quad (1)$$

where $H_{t-k \rightarrow t}$ represents the accumulated camera transformation from frame I_{t-k} to frame I_t . Instead of repeatedly transforming already-warped coordinates, the accumulated homography is applied only once to the detected objects’ centroids. This reduces numerical drift and provides a stable reference for estimating the residual velocity in the next stage.

3.2. Multi-Scale Residual Velocity

We first define three velocities:

- **Raw velocity:** the observed image-plane velocity of an object, computed from the displacement of its centroid between two frames separated by g frames and normalized by the corresponding time interval.
- **Ego-motion-induced velocity:** the image-plane velocity caused by camera motion, computed from the displacement of the object’s location predicted by the homography between two frames separated by g frames and normalized by the corresponding time interval.

- Residual velocity: the camera-motion-compensated object velocity obtained by subtracting the ego-motion-induced velocity from the raw velocity.

Based on the accumulated homography matrix and the objects' bounding boxes, we aim to estimate the *residual velocity*. For a temporal gap g , the raw velocity v_g^{raw} is calculated by dividing the displacement of the object centroid between two frames (I_{t-g} and I_t) by the time gap g . The raw velocity mixes the object's own motion with the camera's motion. To remove the camera motion, the accumulated homography is applied to an object's centroid $\mathbf{o}_{t-g}$ at I_{t-g} , yielding $\hat{\mathbf{o}}_t$, the centroid predicted at I_t under the assumption that the object is static. Then the ego-motion-induced displacement $\hat{\mathbf{o}}_t - \mathbf{o}_{t-g}$ is normalized by the time interval g to estimate the ego-motion-induced velocity v_g^{ego} . The residual velocity v_g^{res} is then computed as

$$v_g^{\text{res}} = v_g^{\text{raw}} - v_g^{\text{ego}}. \quad (2)$$

This value approximates the motion that a static camera would observe. For example, a parked vehicle has near-zero residual velocity even when camera motion causes a large image displacement. All displacements are normalized by the image dimensions to make them independent of the image resolution. In the implementation, the factor $1/g$ is omitted; the difference is absorbed by a per-gap rescaling of the first projection layer of Sec. 3.3.

To capture motion over different time spans, residual velocities are computed at temporal gaps of $g \in \{2, 5, 10\}$ frames. That means three accumulated homographies of Eq. (1) are maintained per frame: $H_{t-2 \rightarrow t}$, $H_{t-5 \rightarrow t}$, and $H_{t-10 \rightarrow t}$. The residual velocities are concatenated with the object's box information to form a 12-dimensional motion feature,

$$(v_2^{\text{res}}, v_5^{\text{res}}, v_{10}^{\text{res}}, \Delta w, \Delta h, c_x, c_y, w, h), \quad (3)$$

where each residual velocity contributes its two image-plane components, and $\Delta w, \Delta h$ capture the change in box scale. The terms c_x and c_y describe the coordinates of the box center, and w and h describe the width and height of the box, respectively.

The largest gap requires 11 contiguous frames, so a track contributes a motion feature only after its history reaches that length. For shorter histories, the module produces no score, and the host score passes through unchanged. The threshold is fixed by the largest gap and adds no free parameter.

3.3. Motion-Language Matching

Given the motion features mentioned above and the language expression, this stage measures how much the observed object's residual motion matches the expression. The 12-dimensional motion feature of Eq. (3) is encoded by a motion encoder, while the language expression is encoded into a 384-dimensional embedding by the pre-trained all-MiniLM-L6-v2 sentence encoder (MiniLM) [17]. Both embeddings are projected into the same feature space using linear projection layers and a shared multilayer perceptron (MLP).

The model is trained using the InfoNCE [18] loss to maximize the similarity between matched motion-expression pairs while pushing unmatched pairs apart. A positive pair is formed by the embeddings of an object and the corresponding expression. Other object-expression pairs in the same mini-batch are treated as negatives. The MiniLM text encoder is kept fixed, and only the projection layers and motion encoder are trained. During inference, the cosine similarity s_{gmc} between an object's motion embedding and the target expression's embedding is used as the matching score and passed to the late fusion stage described below.

3.4. Fusion

We aim to integrate the matching score described above with the prediction score from the host RMOT method to improve tracking performance. The integration is defined as

$$s_{\text{final}} = s_{\text{host}} + \alpha_c s_{\text{gmc}}, \quad (4)$$

where s_{host} is the prediction score from the host model, and the weight α_c depends on the expression class $c \in \{\text{mot}, \text{app}\}$: moving and static expressions receive α_{mot} , and appearance expressions receive α_{app} . For example, "moving car" is a *moving* expression, "left vehicles which are parking"¹ is a *static* expression, and "right cars in light color" is an *appearance* expression. Based on the keywords in the expression, we classify it as moving, static, or appearance.² Because different host models produce scores over different ranges, the weight α_c is selected for each host via leave-one-sequence-out (LOSO) cross-validation. In each fold, one test sequence is held out and the weight that yields the highest pooled HOTA on the remaining sequences is selected. The final weight, applied to the full test set, is the median over folds. For both FlexHook settings, the procedure selects the same weight for all expression classes, so the class-conditional weighting reduces to a single weight.

Since the GMC module only adds a term to the final prediction score, it can be integrated into existing RMOT frameworks, either two-stage or end-to-end query-based, without changing or retraining their detector, tracker, visual encoder, or language encoder.

4. EXPERIMENTS

4.1. Setup

We evaluate the proposed module on the Refer-KITTI V1 [1] and the Refer-KITTI V2 [9] benchmarks using the HOTA metric from TrackEval [19], which jointly measures detection and association performance. We report both moving-class HOTA, which evaluates only the moving-class expressions (those assigned to the moving class by the keyword classifier of Sec. 3.4), and pooled HOTA, which evaluates all expressions in the benchmark. The V1 test split contains three sequences, and the V2 test split contains four.

Refer-KITTI is highly imbalanced: only 21 of the 150 V1 test expressions and 111 of the 862 V2 test expressions are moving-class. Consequently, improvements in motion understanding have only a limited impact on pooled HOTA. Moving-class HOTA is therefore used as the primary evaluation metric, while pooled HOTA is reported to reflect overall benchmark performance.

The proposed GMC module is integrated with three host architectures — iKUN [2], FlexHook [3], and TransRMOT [1] — yielding four host settings, since FlexHook is evaluated on both Refer-KITTI V1 and V2. For comparison, we also evaluate a raw-velocity variant that uses image-plane motion directly, without ego-motion compensation. The GMC module is trained using the AdamW [20] optimizer with a learning rate of 10^{-3} and a batch size of 256, for 100 epochs. The linear projectors and MLP in the module are trained on the Refer-KITTI V1 training split (15 sequences, disjoint from the test

¹In Refer-KITTI, this expression refers to a vehicle that is already parked, rather than a vehicle that is in the process of parking.

²The classifier matches lowercased stems. Moving: moving, in motion, driving, walking, running, jogging, crossing, riding, travelling/traveling, braking, brake, accelerat-, decelerat-, slowing down, speeding up, approaching, overtaking, receding. Static: parking, parked, stopped, stop, stand, static, stationary. All other expressions are appearance. The list was fixed before the fusion weights were selected.

sequences of both benchmarks). The same V1-trained aligner is used unchanged for all four host settings, including the Refer-KITTI V2 evaluation. Each positive pair consists of an expression and the motion features derived from the object’s ground-truth boxes. At test time, the raw velocity is computed from the host model’s initial tracking results, the ego-motion-induced component is subtracted as in Eq. (2), and the resulting residual velocities form the motion features. We track up to 600 corners per frame with a 3-pixel RANSAC threshold; the road-plane homography is successfully estimated for all 7,690 adjacent frame pairs of the 19 training and evaluation sequences (the 18 Refer-KITTI V1 sequences plus the one V2-only test sequence). Excluding host inference, the module runs at 149 FPS on CPU (6.7 ms per frame).

Results are averaged over three random seeds for the main configurations; reported $\pm$ values are standard deviations over seeds. Ablation analyses are done using five seeds together with Welch’s t -test. All host models use their original detection results, and every Refer-KITTI V1 setting is evaluated on the benchmark’s official 150-expression test list. Our reproduced baselines match the published FlexHook scores at their reported precision (V1 53.824 vs. 53.83, V2 42.526 vs. 42.53), while our iKUN reproduction scores 44.543 against the published 44.56. TransRMOT’s published scores (46.56 before and 38.06 after the community frame-numbering correction) use different ground-truth conventions and are not directly comparable; Table 3 therefore lists our reproduction under the host’s own frame convention. In every setting, the fused model reproduces our reproduced baseline exactly at $\alpha_c=0$.

Refer-KITTI does not provide a dedicated validation split. All fusion weights are therefore selected by the leave-one-sequence-out procedure of Sec. 3.4: within each fold, the held-out sequence does not participate in the selection, and the reported models apply the median weight over folds to the full test set. As a check, an out-of-fold composite on an earlier configuration of the module, in which each sequence is scored under weights fitted with that sequence held out, retained approximately 88% of the moving-class gain. The selected weights for the four host settings are listed in Table 1. Their magnitudes are not comparable across hosts because the hosts’ score scales differ: the standard deviation of the host score is 0.50 for iKUN and 0.35 for TransRMOT, but 10.2 for FlexHook’s score margins. The detection threshold of each host is kept at its native value (0.0 for iKUN and FlexHook; 0.5 for TransRMOT), so the module only re-ranks the candidates the host already emits.

4.2. Main Results

Table 2 reports HOTA on the moving-class expressions, the primary metric. The GMC module improves all four settings. The gains differ by more than a factor of twenty, and they decrease as the host’s baseline moving-class HOTA increases — the largest gain on the weakest host, the smallest on the strongest — across all four settings. This ordering is consistent with the compensated motion cues supplying what the host’s own motion understanding lacks, although the four baselines differ in detector, benchmark, and ground-truth convention, so part of it may reflect the available headroom.

Table 1. Fusion weights α_c selected by leave-one-sequence-out cross-validation.

Host	α_{mot} (moving/static)	α_{app} (appearance)
iKUN	1.0	0.1
TransRMOT	0.7	0.5
FlexHook V1	7.0	7.0
FlexHook V2	5.0	5.0

Table 2. Moving-class HOTA (moving-class expressions: 21/150 for iKUN, TransRMOT, and FlexHook V1; 111/862 for FlexHook V2). Performance gain = fused – baseline moving-class HOTA (mean $\pm$ standard deviation over $n=3$ seeds). Rows are ordered by baseline moving-class HOTA.

Host	Baseline	Performance gain
iKUN	27.70	+9.44 $\pm$ 0.92
TransRMOT	33.88	+1.43 $\pm$ 0.13
FlexHook V2	38.56	+0.69 $\pm$ 0.04
FlexHook V1	47.90	+0.43 $\pm$ 0.19

Table 3. Pooled HOTA, DetA, and AssA on the Refer-KITTI test sets. Each host row gives the baseline; the + GMC row gives the fused model (mean $\pm$ standard deviation over three seeds). * denotes our reproduction using the official implementation (the FlexHook reproductions match the published 53.83/42.53 at their reported precision; TransRMOT is evaluated under the host’s own frame convention, Sec. 4.1). Baseline DetA and AssA are from our reproductions.

Method	HOTA	DetA	AssA
<i>Refer-KITTI</i>			
iKUN [2]	44.56	32.04	62.48
+ GMC	45.30 $\pm$ 0.12	32.90	62.94
TransRMOT [1]	45.76*	36.64	57.40
+ GMC	46.15 $\pm$ 0.03	36.95	57.89
FlexHook V1 [3]	53.82*	43.35	66.92
+ GMC	53.98 $\pm$ 0.06	43.51	67.07
<i>Refer-KITTI-V2</i>			
FlexHook V2 [3]	42.53*	30.63	59.19
+ GMC	42.63 $\pm$ 0.03	30.75	59.23

Table 3 summarizes the pooled HOTA results with and without the GMC module. The proposed module improves every setting. For iKUN and both FlexHook settings, the fused model also exceeds the host’s published score; for TransRMOT, the comparison is against our reproduction (Sec. 4.1).

4.3. Ablation Studies

Table 4 reports the five-seed ablation measurements when iKUN is used as the host model. The full module improves moving-class HOTA from 27.70 to 36.99 and pooled HOTA from 44.54 to 45.28 (five-seed means; the three-seed means reported in Sec. 4.2 are 37.14 and 45.30, and 44.54 is our reproduction of the published 44.56). If the ego-motion compensation is removed, the moving-class HOTA drops from 36.99 to 32.37, while removing the multi-scale gaps when calculating residual velocity reduces it from 36.99 to 35.14. Both drops are significant under Welch’s t -test between the five seeds of the full module and the five seeds of each variant (a seed-stability test on the fixed test split): removing the ego-motion compensation gives $t=13.1$ ($df=6.3$, $p=8\times 10^{-6}$) on moving-class and $t=13.2$ ($df=7.5$, $p=2\times 10^{-6}$) on pooled HOTA; removing the multi-scale gaps gives $t=5.3$ ($df=6.3$) and $t=6.8$ ($df=4.5$), both $p=1.6\times 10^{-3}$. The class-conditional weighting itself carries part of the headline gain: with a single weight, the moving-class HOTA reaches 32.42, so the routing contributes +4.57 and the ego-motion compensation +4.62 of the total +9.29; pooled HOTA, which the routing barely moves, serves as the routing-independent check. HOTA on the expressions outside the moving class changes little across the variants (Others in Table 4). Both components therefore contribute to the gain.

Figure 2 shows a sample tracking result. The parked car appears to move across frames due to camera motion, while the moving car,

Table 4. Per-class HOTA ablation on iKUN ($n=5$ seeds). The full module and the $-ego / -multiscale$ variants all use the weights of Table 1 ($\alpha_{mot}=1.0$, $\alpha_{app}=0.1$); the variants inherit them without re-selection. The single- α row uses $\alpha_{mot}=\alpha_{app}=0.35$, the single-weight LOSO optimum. Others = all expressions outside the moving class.

Variant	Moving HOTA	Others HOTA	Pooled HOTA
Native host (w/o GMC)	27.70	45.87	44.54
Full module (w. GMC)	36.99	46.05	45.28
single α (no routing)	32.42	45.99	44.95
$- ego$ (raw velocity only)	32.37	45.77	44.61
$- multiscale$ (single gap)	35.14	45.96	45.00



Fig. 2. “Moving cars” (Refer-KITTI seq. 0005; iKUN host). The parked car (red) sweeps across the image as the camera passes, while the moving car (green) barely shifts in the image plane. In this example, raw image motion misleads the matching for both cars; the ego-compensated residual velocity recovers both.

keeping nearly the same distance from the camera, barely shifts in the image. Without ego-motion compensation, the two cases are easily confused, and the matching scores follow the apparent motion. After motion compensation, the parked car has nearly zero residual velocity, while the moving car has a clearly nonzero residual velocity. This allows the GMC module to rank the two cars correctly.

5. CONCLUSION AND LIMITATIONS

The GMC module is a plug-in that improves motion grounding in referring multi-object tracking without modifying the detector, tracker, or appearance-language matching model. By compensating for camera motion, estimated from road-region background correspondences, and matching the residual motion features against language expressions, the module yields a matching score that is integrated with the host model’s prediction via late fusion, helping distinguish object motion from camera-induced motion. On the Refer-KITTI benchmarks, pooled HOTA improves in all four host settings, and the moving-class gain grows as the host’s own motion understanding weakens, with

the largest gain observed on iKUN ($+9.44 \pm 0.92$).

Despite these promising results, several limitations remain. First, the road-plane homography assumes a flat ground plane. Scenes with steep slopes or uneven terrain fall outside the model’s scope. Second, the keyword classifier routes only moving- and static-class expressions to α_{mot} ; direction, turning, and relative-speed expressions (24 of the 150 V1 test expressions; 214 of the full 818-expression V1 set) receive the appearance weight, so the module does not help them even though they describe motion. Third, the module targets hosts without a native temporal motion memory; in our preliminary experiments, cascading it onto a temporally-aware host (TempRMOT [9]) degraded HOTA, so we regard such hosts as out of scope. Finally, we evaluated three RMOT frameworks but only on the Refer-KITTI benchmarks. Additional evaluations on other datasets are needed to further validate the generality of the proposed approach.

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